EE2012 Midterm Cheat Sheet

By: Brians Tjipto

1 Set Operations

Operation	Meaning
$A \subseteq B$	A subset of B
$A \cap B$ or AB	Intersection
$A \cup B$	Union
A^c or $\bar{A}$	Complement
$\emptyset$	Empty set
S	Sample space

Identities:

$$A \cup B = B \cup A$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$(A \cup B)^c = A^c \cap B^c$$

2 Probability Axioms

- $P(A) \geq 0$
- $P(S) = 1$
- $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$

Properties:

$$P(\emptyset) = 0$$
$$P(A^c) = 1 - P(A)$$
$$P(A) \leq 1$$
$$A \subseteq B \Rightarrow P(A) \leq P(B)$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

3 Conditional Probability

Definition: $P(A|B) = \frac{P(A \cap B)}{P(B)}$, $P(B) > 0$ **Multiplication Rule:** $P(A \cap B) = P(A|B)P(B)$ **Chain Rule:** $P(A \cap B \cap C) = P(A)P(B|A)P(C|AB)$
Independence: $A \perp B \iff P(A \cap B) = P(A)P(B)$

- $\iff P(A|B) = P(A)$ (if $P(B) > 0$)
- $\iff P(B|A) = P(B)$ (if $P(A) > 0$)

Total Probability: $P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$ **Bayes' Rule:** $P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$ **Event A before B:** $P(\text{A before B}) = \frac{P(A)}{P(A)+P(B)}$

4 Counting Methods

4.1 How to Choose a Counting Method

- **Does order matter?**
  - **YES** (Permutations - passwords, rankings)
    - **Repeatable? YES**: n^k. **NO**: $P(n, k)$.
  - **NO** (Combinations - lottery, hands)
    - **Repeatable? YES**: $\binom{n+k-1}{k}$. **NO**: $\binom{n}{k}$.

Key Principles:

- **Sum Rule (Cases):** For disjoint (mutually exclusive) options, the total number of choices is the sum of choices for each option.
- **Inclusion-Exclusion (Counts):** $|A \cup B| = |A| + |B| - |A \cap B|$.

Type	Formula
Permutation	$P(n, k) = \frac{n!}{(n-k)!}$
Combination	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$
With replacement	n^k
Circular	$(n-1)!$
Stars/Bars	$\binom{n+k-1}{k}$
Multinomial	$\binom{n}{n_1, \ldots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$

5 Random Variables

PMF: $p_X(x) = P(X = x)$. Properties: $p_X(x) \geq 0$, $\sum_x p_X(x) = 1$.
CDF: $F_X(x) = P(X \leq x)$. Properties: Non-decreasing, right-continuous. **Probability via CDF:**

- $P(X \leq a) = F_X(a)$
- $P(X > a) = 1 - F_X(a)$
- $P(a < X \leq b) = F_X(b) - F_X(a)$
- $P(X = a) = F_X(a) - F_X(a^-)$

5.1 PMF of a Function of RVs

To find PMF of $Z = g(X, Y)$: $P(Z = z) = \sum_{(x,y)|g(x,y)=z} P(X = x, Y = y)$. **Ex (Sum of 2 fair dice, $Z = X + Y$):** $P(Z = 4) = P(1, 3) + P(2, 2) + P(3, 1) = \frac{3}{36}$.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

6 Expectation & Variance

Expectation: $E[X] = \sum_x x \cdot p_X(x)$

$$E[g(X)] = \sum_x g(x) \cdot p_X(x)$$

Variance: $\mathrm{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$
$\sigma_X = \sqrt{\mathrm{Var}(X)}$
Properties:

$$E[aX + b] = aE[X] + b$$
$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$$
$$E[X + Y] = E[X] + E[Y]$$
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ (if indep.)}$$

7 Discrete Distributions

Distr.	PMF (k successes, trial n)	Mean	Var.
Bernoulli	$p^k(1-p)^{1-k}$	p	$p(1-p)$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Geometric	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson	$\frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ
Neg. Binom.	$\binom{n-1}{k-1} p^k (1-p)^{n-k}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$
Uniform	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$

8 Useful Formulas

Binomial Theorem: $(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$
Series: $\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$; $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$
Combinations: $\sum_{k=0}^{n} \binom{n}{k} = 2^n$
The number of heads with the highest probability (the mode) is given by $k = \lfloor (n+1)p \rfloor$

9 Mock Exam Walkthroughs

9.1 AY24-25 Midterm Solutions

Q1.1.a: A and B sit together

Concept: Permutations (block method). **1)** Total arrangements of 6 people: $6! = 720$. **2)** Block (A,B), arrange 5 items ((AB),C,D,E,F): $5! = 120$. **3)** Internal swaps in (AB): $2! = 2$. **4)** Favorable: $5! \times 2! = 240$. **5)** Prob: $\frac{240}{720} = \frac{1}{3}$.

Q1.1.b: A, B not together AND E, F not together

Concept: Inclusion-Exclusion. Let $S_1 = \{\text{A,B together}\}, S_2 = \{\text{E,F together}\}$. Find $P(S_1^c \cap S_2^c) = 1 - [P(S_1) + P(S_2) - P(S_1 \cap S_2)]$. **1)** $|S_1| = |S_2| = 240$. **2)** $|S_1 \cap S_2|$: Block (AB), Block (EF). Arrange 4 items ((AB),(EF),C,D): $4! = 24$. Internal swaps: $2!$ for A,B and $2!$ for E,F. Total: $4! \times 2! \times 2! = 96$. **3)** Favorable: $720 - 240 - 240 + 96 = 336$. **4)** Prob: $\frac{336}{720} = \frac{7}{15}$.

Q1.2.a: Conditional Binomial

Concept: Counting with Combinations. Given 4H in 6 flips, find $P(\text{first two are H})$. **1)** Total ways for 4H in 6 flips: $\binom{6}{4} = 15$. **2)** Favorable ways: HH____. Need 2H in last 4 spots: $\binom{4}{2} = 6$. **3)** Prob: $\frac{\text{Favorable}}{\text{Total}} = \frac{6}{15} = \frac{2}{5}$.

Q1.2.b: Find Binomial parameter p

Concept: Binomial PMF algebra. Given $P(X = 2) = 4P(X = 4)$ for $n = 6$. **1)** Setup: $\binom{6}{2}p^2(1-p)^4 = 4\binom{6}{4}p^4(1-p)^2$. **2)** Since $\binom{6}{2} = \binom{6}{4} = 15$, cancel terms to get $(1-p)^2 = 4p^2$. **3)** Solve: $1 - p = 2p \Rightarrow 3p = 1 \Rightarrow p = 1/3$. **4)** Find $P(X = 4) = \binom{6}{4}(\frac{1}{3})^4(\frac{2}{3})^2 = 15 \cdot \frac{4}{729} = \frac{20}{243}$.

Q1.3.a: Max of two dice

Concept: PMF of a function of RVs. Find $P(\max(X, Y) = 5)$. **1)** Total outcomes: $6 \times 6 = 36$. **2)** Favorable pairs: (5,1-5) and (1-4,5). Total 9 pairs. **3)** Prob: $\frac{9}{36} = \frac{1}{4}$.

Q1.3.b: A wins game of turns

Concept: Recursive Probability. **1)** Let W be prob A wins. Let p be prob of success (sum=10). Pairs are (4,6),(5,5),(6,4), so $p = 3/36 = 1/12$. **2)** A wins if A succeeds (prob p), OR A fails $(1-p)$, B fails $(1-p)$, and it is A's turn again (prob W). **3)** Equation: $W = p + (1-p)^2 W$. **4)** Solve: $W = \frac{1}{12} + (\frac{11}{12})^2 W \Rightarrow W(1 - \frac{121}{144}) = \frac{1}{12} \Rightarrow W(\frac{23}{144}) = \frac{1}{12} \Rightarrow W = \frac{12}{23}$.

Q2.a: Picking marbles

Concept: Independent events. **1)** Picking from {R,G,B} with equal prob $\implies P(R) = 1/3$ for each independent pick. **2)** $P(\text{all 3 R}) = P(R)^3 = (1/3)^3 = 1/27$.

Q2.b: Bayes' Rule with marbles

Concept: Bayes' & Total Probability. Find $P(B|A)$ for $B = \{\text{all 3 R}\}, A = \{\text{2 picked are R}\}$. **1)** Bayes': $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$. We know $P(B) = 1/27$ and $P(A|B) = 1$. **2)** Find $P(A)$ with Total Prob over number of red marbles in bag, R_i. Bag composition is Binomial $N_R \sim Bin(3, 1/3)$. **3)** $P(A) = P(A|R_2)P(R_2) + P(A|R_3)P(R_3)$. **4)** $P(R_2) = \binom{3}{2}(\frac{1}{3})^2(\frac{2}{3}) = \frac{6}{27}$. $P(R_3) = \frac{1}{27}$. **5)** $P(A|R_2) = \binom{2}{2}/\binom{3}{2} = 1/3$. $P(A|R_3) = 1$. **6)** $P(A) = (\frac{1}{3})(\frac{6}{27}) + (1)(\frac{1}{27}) = \frac{3}{27} = \frac{1}{9}$. **7)** $P(B|A) = \frac{1 \times 1/27}{1/9} = 1/3$.

Q3.a: Find constant in PMF

Concept: $\sum p_X(x) = 1$. **1)** Sum: $p + 2p + 3p + 4p = 10p = 1 \Rightarrow p = 0.1$.

Q3.b: Find E[X

and E[X^2]] **Concept:** Expectation definition. **1)** $E[X] = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$. **2)** $E[X^2] = 1^2(0.1) + 2^2(0.2) + 3^2(0.3) + 4^2(0.4) = 10$.

Q3.c: Find Var[1-2X

] **Concept:** $Var(aX + b) = a^2 Var(X)$. **1)** $Var(X) = E[X^2] - (E[X])^2 = 10 - 3^2 = 1$. **2)** $Var(1 - 2X) = (-2)^2 Var(X) = 4(1) = 4$.

9.2 AY23-24 Midterm Solutions

Q1.a: Binomial Probability

Concept: Binomial. Fair coin $p = 0.5, n = 10$, find $P(X = 2)$. **1)** Formula: $P(X = 2) = \binom{10}{2}(0.5)^{10} = 45/1024$.

Q1.b: Find n from Binomial Property

Concept: Binomial property $\binom{n}{k} = \binom{n}{n-k}$. **1)** Given $P(X = 3) = P(X = 5)$ for fair coin, we get $\binom{n}{3} = \binom{n}{5}$. **2)** This means $n - 3 = 5 \Rightarrow n = 8$. **3)** Find $P(2H)$ for $n = 8$: $\binom{8}{2}(0.5)^8 = 28/256 = 7/64$.

Q1.c: Conditional Binomial

Concept: Binomial and Counting. **1)** $P(3H) = \binom{5}{3}(0.8)^3(0.2)^2 = 0.2048$. **2)** For conditional part, count ways: Total ways for 3H is $\binom{5}{3} = 10$. Favorable ways (H___H) means 1H in 3 middle spots: $\binom{3}{1} = 3$. Prob: $3/10 = 0.3$.

Q1.d: Geometric then Binomial

.**Concept:** Geometric to find p, then find Binomial mode. **1)** $P(X \geq 3) = 0.16$ for $Geom(p)$ means first two are tails: $(1-p)^2 = 0.16 \Rightarrow 1 - p = 0.4 \Rightarrow p = 0.6$. **2)** For $Y \sim Bin(10, 0.6)$, mode is $\lfloor (n+1)p \rfloor = \lfloor 11 \cdot 0.6 \rfloor = 6$. **3)** $P(Y = 6) = \binom{10}{6}(0.6)^6(0.4)^4 \approx 0.2508$.

Q1.e: 5 heads before 3 tails

Concept: Negative Binomial logic. **1)** Game ends within $5+3-1 = 7$ flips. **2)** "5H before 3T" is same as "$\geq$ 5 heads in 7 flips". **3)** Let $X \sim Bin(7, 0.8)$. Find $P(X \geq 5) = P(5) + P(6) + P(7)$. **4)** Summing the binomial probabilities gives ≈ 0.8520.

Q2: 3 Coins, Total Prob & Bayes'

Concept: Total Prob & Bayes'. C_i=Coin i, A=3H, B=2H in first 2, $C = A \cap B$. **(a)** $P(B|A)$: By counting, given 3H in 5 flips ($\binom{5}{3} = 10$ ways), ways to have HH in first two is $\binom{3}{1} = 3$. So $P = 3/10$. **(b)** $P(A) = \frac{1}{3}\sum P(A|C_i) = \frac{1}{3}[\binom{5}{3}(\frac{1}{2})^5 + \binom{5}{3}(\frac{2}{3})^3(\frac{1}{3})^2 + 0] \approx 0.2139$. **(c)** $P(C_1|C) = \frac{P(C|C_1)P(C_1)}{P(C)}$. Event C is HH in first 2, 1H in last 3. $P(C|C_1) = (\frac{1}{2})^2\binom{3}{1}(\frac{1}{2})^3 = 3/32$. $P(C|C_2) = (\frac{2}{3})^2\binom{3}{1}(\frac{2}{3})^1(\frac{1}{3})^2 = 24/243$. $P(C) = \frac{1}{3}[3/32 + 24/243] \approx 0.0642$. $P(C_1|C) = \frac{(3/32)(1/3)}{0.0642} \approx 0.487$.

Q3.a: Expectation of Dice Sum

Concept: PMF and Expectation. Two 4-sided dice. **1)** $E[X] = E[D_1] + E[D_2]$. $E[D_1] = (1 + 4)/2 = 2.5$. So $E[X] = 2.5 + 2.5 = 5$.

Q3.b: Binomial with Dice Difference

Concept: Find event prob, then Binomial. **1)** $Y = |D_1 - D_2|$. Find $p = P(Y \leq 1)$. $Y = 0$ has 4 outcomes. $Y = 1$ has 6 outcomes. Total 10 outcomes. $p = 10/16 = 5/8$. **2)** $P(\text{3 successes in 10 rolls}) = \binom{10}{3}(\frac{5}{8})^3(\frac{3}{8})^7 \approx 0.0306$.

Q3.c: Event A before Event B

Concept: $P(A \text{ before } B) = \frac{P(A)}{P(A)+P(B)}$. **1)** $A = \{Y \leq 1\}, B = \{Y > 2\}$. $P(A) = 10/16$. **2)** $Y > 2 \Rightarrow Y = 3$. Outcomes are (1,4), (4,1), so $P(B) = 2/16$. **3)** Prob: $\frac{10/16}{10/16+2/16} = \frac{10}{12} = \frac{5}{6}$.

Found: https://github.com/brianstm/NUS.git